Mining Site Recommendation

Team B5

Context

- Copper-nicus Mining LTD. is looking to develop a **new mining site** in British Columbia's Nicus Mountain range.
- Three potential sites have been located, each connected to nearby **water sources**, existing **infrastructure** and **First Nations communities.**
- Our goal is to recommend which proposed site should be developed and the mining strategy to use.

Proposed Sites

Figure 1: Map of Proposed Deposit Sites and Nearby Features

Deposit 1

- Deep underground deposit, high grade ore.
- Highly remote, infrastructure development needed
- Risks surround wildlife migration

Deposit 2

- Increasing ore grade with depth.
- Some additional infrastructure needed.
- water contamination risks.

Deposit 3

- Shallow mine, low grade ore, minimal waste rock
- Usable nearby infrastructure (highway and town)
- Risks surrounding nearby archaeological site.

Overview

Deposit Research

- *Stakeholder consultation*
- *SLCAs*

Decision Making Process

- *WDM*
- *Sensitivity analysis*

Our Decision

- *Deposit selection*
- *Mining strategy*
- *Impacts*

Conclusion

- *Q&A*

Stakeholder Needs & Evaluation Criteria

Key Stakeholders

- Policy Makers (Regulators) (35%)
- Local Indigenous Communities (Rights Holders) (25%)
- Mining Company (20%)
- Local Communities (15%)
- Construction Company (5%)

Stakeholder Needs

- Employment & Economic Growth
- Preservation of Territory
- Respectful Operations
- Profits of Mining Company
- Augment Infrastructure
- Law - Abiding (EGBC)

Evaluation Criteria

- Community Benefits
- Environmental Preservation
- Cultural Site Preservation
- Profitability
- Local Infrastructure Improvements
- Resiliency

Scoring Criteria

Community Benefits

Score 0 → No community benefits

Score 10 → Major community benefits

Environment Preservation

Score 0 → Little to no preservation

Score 10 → No damage on environment

Cultural Sites Preservation

Score 0 → Little to no preservation

Score 10 → No damage to cultural sites

Profitable Mine

Score 0 → $ 4 Billion

Score 10 → $ 8 Billion

Infrastructure Improvement

Score 0 → No infrastructure developed

Score 10 → Major infrastructure improvements

Resiliency

Score 0 → Lowest resiliency

Score 10 → Highest resiliency

Based on six evaluation criteria, Deposit 2 outperforms.

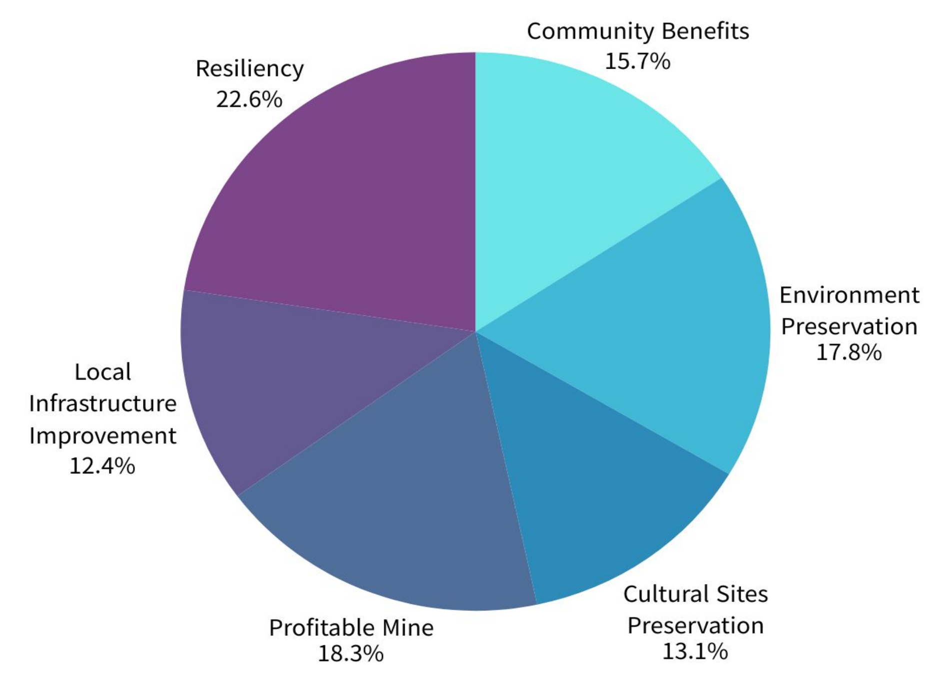


Figure 2: Weight Distribution of WDM

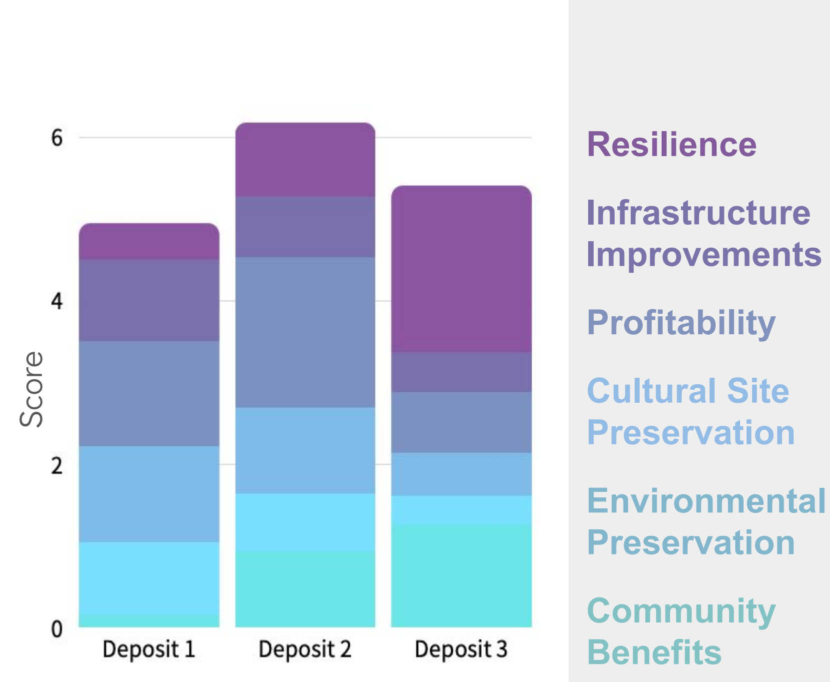
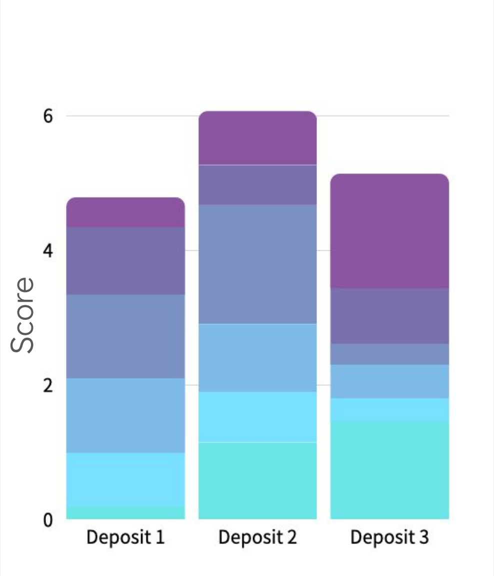
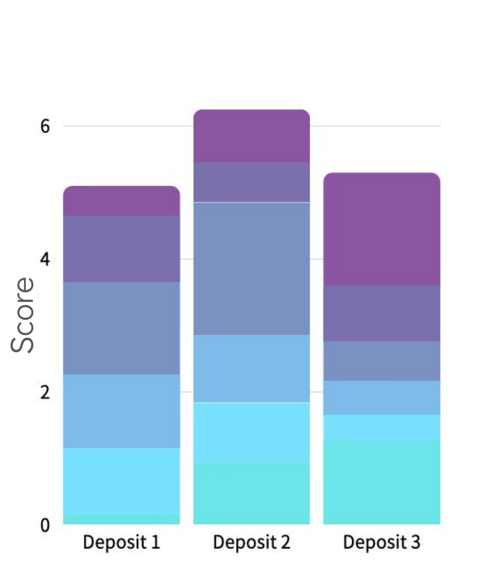


Figure 3: Deposit 2 outperforms other deposits

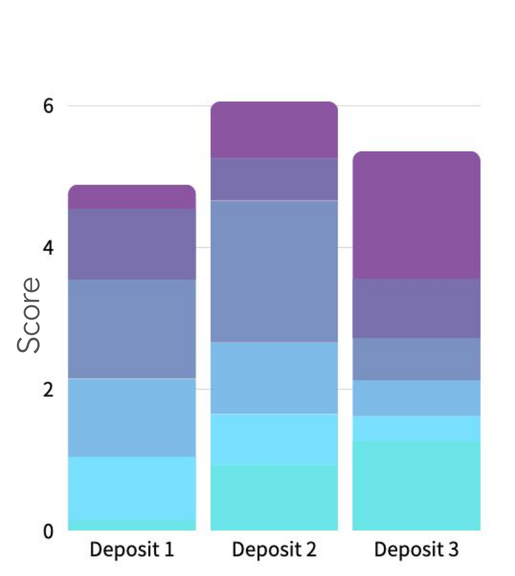
Sensitivity Analysis



Adjusted Weighting of Community Benefits & Profitability



Adjusted Weighting of Environment Preservation & Profitability



Adjusted Weighting of Infrastructure Improvements & Profitability

Resilience

Infrastructure Improvements

Profitability

Cultural Site Preservation

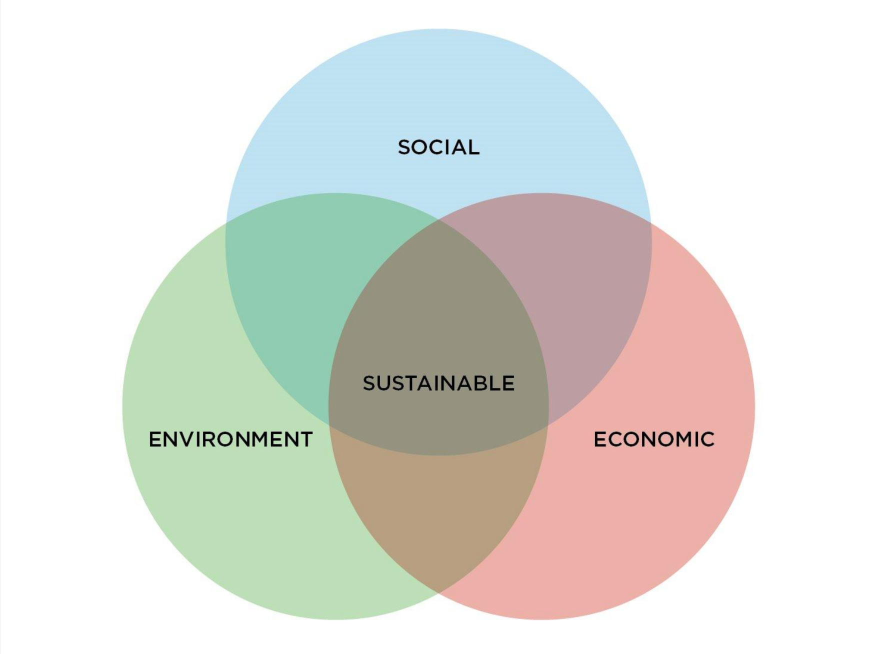
Environmental Preservation

Community Benefits

Further Justification

All aspects of **sustainability** considered:

- Environment:
  - Smaller surface footprint
  - Less material removal
- Social:
  - Culturally site protection
  - Infrastructure improvement
  - Employment opportunities
- Economic:
  - Strong profitability potential
  - Long-term economic viability



Positive Impacts

✓ **Economic Development**

- Generate long-term revenue from copper production
- Support local businesses and economic activities

✓ **Community Benefits**

- Creates employment opportunities
- Supports development of local infrastructure
- Reduced risk to culturally significant sites

✓ **Supporting Future Energy**

- Copper is essential for renewable energy technologies
- Supports the growing demand for electrification

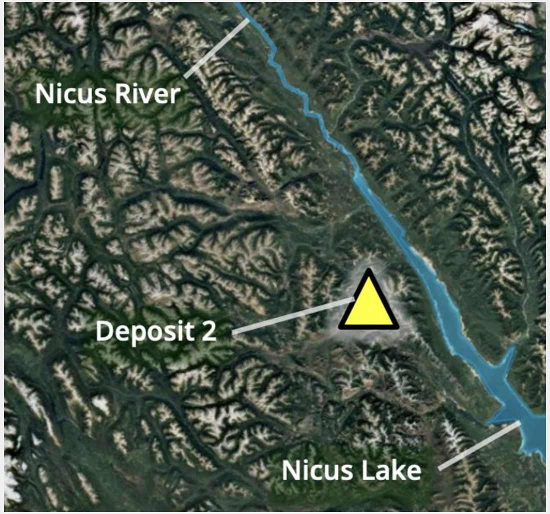
Potential Challenges

✘ **Environmental Risks**

- Close proximity to Nicus Lake and Nicus River
- Risk of water contamination if tailings are not properly managed
- GHG emissions from heavy mining equipment

✘ **Operational Challenges**

- Decreasing rock stability at greater depth
- Economic uncertainty around break-even point



Summary of our Decision

- **Recommended Mining Strategy: Underground Mining at Deposit 2**
- **Key Reasons:**

  Accounting for stakeholder salience

  Strong profitability potential

  Community benefits (jobs & infrastructure)

  Reduced surface disturbance
- **Imperfect, but strongest overall option**

  Environmental Risks

  Operational Challenges

Q&A

Appendices

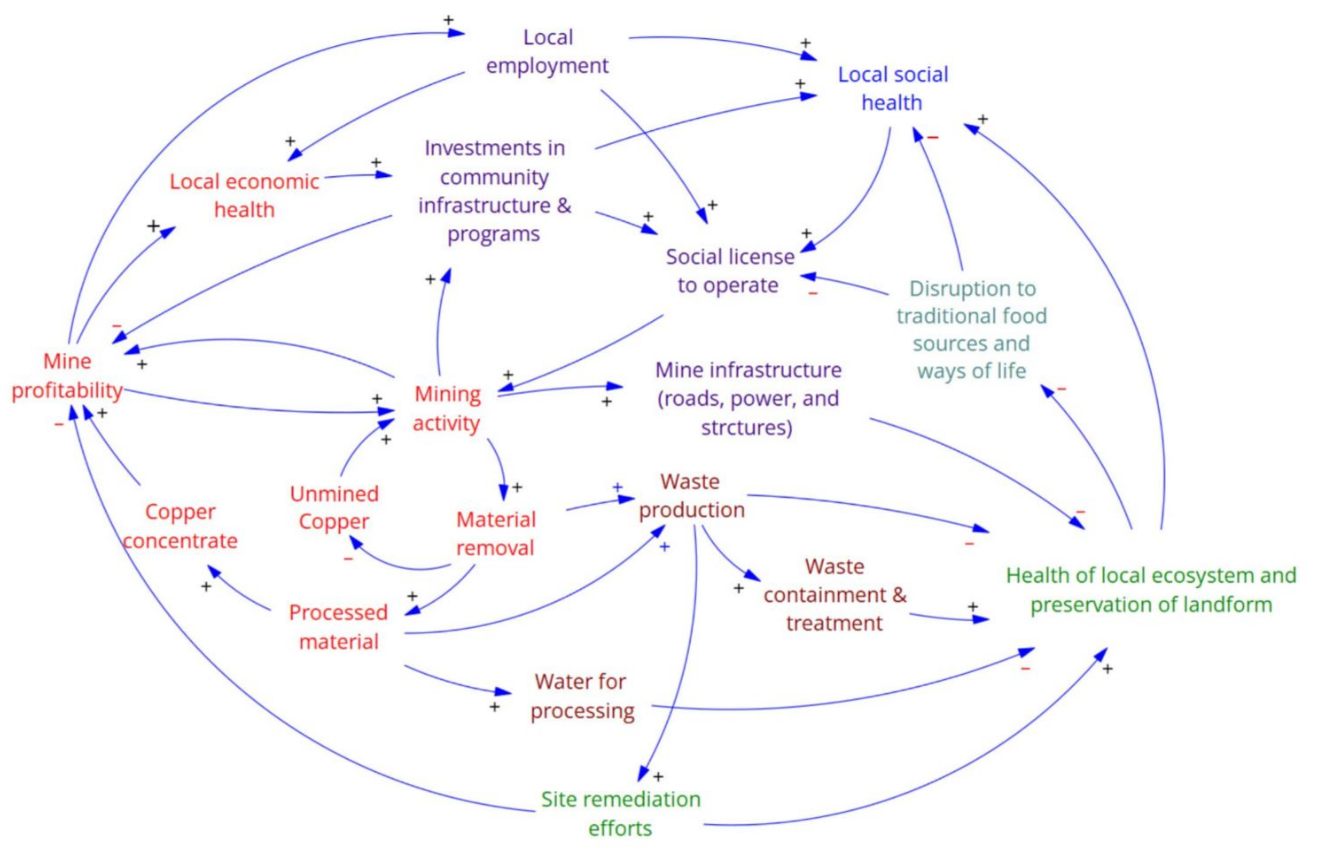
Appendix A - Stakeholder WDM

Stakeholders	Stakeholder Weight	Community benefits		Preservation of the environment		Preservation of cultural sites		Profitability		Improvement to local infrastructure		Resilience	
		Score	Weighted	Score	Weighted	Score	Weighted	Score	Weighted	Score	Weighted	Score	Weighted
Local Communities	15%	10	1.5	8	1.2	4	0.6	9	1.35	10	1.5	9	1.35
Local Indigenous Communities (Rights Holders)	25%	10	2.5	10	2.5	10	2.5	9	2.25	8	2	10	2.5
Policy Makers	35%	5	1.75	7	2.45	5	1.75	3	1.05	1	0.35	7	2.45
Mining Company	20%	1	0.2	3	0.6	1	0.2	10	2	3	0.6	10	2
Construction Company	5%	2	0.1	2	0.1	0	0	8	0.4	6	0.3	8	0.4
Totals:			6.05		6.85		5.05		7.05		4.75		8.7
Weightings for each criteria			15.7%		17.8%		13.1%		18.3%		12.4%		22.6%

Appendix A - Main WDM

Evaluation Criteria	Weight	Deposit 1 (underground)		Deposit 2 (underground)		Deposit 3 (open pit)	
		Score	Weighted	Score	Weighted	Score	Weighted
Community benefits	15.7%	1	0.16	6	0.94	8	1.26
Preservation of the environment	17.8%	5	0.89	4	0.71	2	0.36
Preservation of cultural sites	13.1%	9	1.18	8	1.05	4	0.53
Profitability	18.3%	7	1.28	10	1.83	4	0.73
Improvement to local infrastructure	12.4%	8	0.99	6	0.74	4	0.49
Resilience	22.6%	2	0.45	4	0.91	9	2.04
Totals:			4.95		6.19		5.40

Appendix B - Causal Loop Diagram



There was no use of GenAI for any purposes in this project or presentation.